TURKISH AIRLINES FLIGHT 981

Causation, Liability, Damages,
and Ethical Responsibility

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I. Introduction

On March 3, 1974, Turkish Airlines Flight 981 crashed in France after its rear cargo door opened in flight. The sudden loss of pressure damaged the cabin floor and flight controls. All 346 people aboard died. The aircraft was a McDonnell Douglas DC-10 traveling from Paris to London after an earlier flight from Istanbul.[1]

The central question is why a door failure was allowed to become a disaster. The danger was already known. A similar accident had happened to an American Airlines DC-10 in 1972, but that crew managed to land. Investigators recommended changes to the door-locking system and cabin pressure relief.[2] Flight 981 showed that the corrections had not fully solved the problem.[3]

This report argues that the strongest case concerns McDonnell Douglas, General Dynamics through its Convair division, and Turkish Airlines. Their roles were different: designing and building the aircraft, correcting known defects, and maintaining and checking the aircraft before flight. The FAA also deserves examination for its oversight, although suing the government raises separate legal obstacles.

The report follows the assignment's five questions:

1. Who could sue, who could be sued, and how did the defendants contribute to the crash?
2. What events led to the accident?
3. Which United States court would be the best place to bring the case?
4. How could compensation and responsibility be calculated?
5. What moral and ethical duties did the defendants have?

The historical discussion draws on official accident reports, congressional findings, and published court decisions. The damages section is a classroom estimate in 1974 dollars, not a statement of what the victims' families actually received. It uses one example to show the calculation; real families obviously had different financial circumstances and losses.

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[1] COMMISSION OF INQUIRY, *Report on the Accident to Turkish Airlines DC-10 TC-JAV on 3 March 1974, Aircraft Accident Report 8/76*, at 1, 4–6, 47–48 (1976) (English translation), https://assets.publishing.service.gov.uk/media/5422eedde5274a1317000247/8-1976_TC-JAV.pdf.

[2] NAT'L TRANSP. SAFETY BD., NTSB-AAR-73-2, *Aircraft Accident Report: American Airlines, Inc., McDonnell Douglas DC-10-10, N103AA, Near Windsor, Ontario, Canada, June 12, 1972*, at Synopsis, 1; app. G, 37–38 (1973), https://www.ntsb.gov/investigations/AccidentReports/Reports/AAR7302.pdf.

[3] COMMISSION OF INQUIRY, *supra* note 1, at 46–49.

II. Facts and Timeline

II.A. Warnings before the crash

The accident's history began years before Flight 981 took off. McDonnell Douglas was responsible for the complete DC-10, while Convair worked on the fuselage and cargo doors. This division of work meant that both companies had information relevant to the aircraft's safety.[4]

Table 1. Warnings Before the Crash

Date	Event
1969	Convair prepared an analysis identifying the danger of a door appearing secure when it was not. It also discussed sudden decompression and damage to the cabin floor.
May 29, 1970	A forward cargo door opened during a ground pressure test and damaged the floor nearby.
June 12, 1972	American Airlines Flight 96 lost its rear cargo door in flight near Windsor, Ontario. The crew landed safely despite damage to the floor and controls.
June 27, 1972	A Convair engineer's memorandum criticized the door design and the proposed response to its problems.
July 6, 1972	The NTSB recommended improvements to locking and pressure relief.
March 3, 1974	Flight 981 suffered a rear cargo-door failure, followed by a fatal loss of control.

Sources:[5,6,7].

The 1970 failure involved a forward door on the ground; the later accidents involved rear doors in flight. Together, they showed that a cargo-door failure could damage more than the door itself.

For the plaintiffs, this history would help establish **notice**: the companies had reasons to recognize the danger before the fatal crash. It would be harder to describe Flight 981 as an unforeseeable accident when a similar failure had already happened in service.

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[4] SPECIAL SUBCOMM. ON INVESTIGATIONS OF THE H. COMM. ON INTERSTATE & FOREIGN COM., 93D CONG., *Air Safety: Selected Review of FAA Performance*, at 11–14 (Comm. Print 1974), https://www.govinfo.gov/content/pkg/CPRT-93HPRT43720O/pdf/CPRT-93HPRT43720O.pdf.

[5] SPECIAL SUBCOMM. ON INVESTIGATIONS, *supra* note 4, at 11–14.

[6] NAT'L TRANSP. SAFETY BD., *supra* note 2, at app. G.

[7] COMMISSION OF INQUIRY, *supra* note 1, at 47–48.

II.B. The missed opportunity in 1972

American Airlines Flight 96 was the clearest warning. Its rear cargo door separated, the cabin floor was damaged, and some controls were affected. The crew was able to land. The NTSB reported minor injuries to two flight attendants and nine passengers.[8]

The NTSB recommended two main protections. First, the handle and small vent door should not reach their normal closed positions until the lock pins were engaged. Second, pressure-relief vents should prevent a large pressure difference from damaging the cabin floor. One measure addressed the door opening; the other addressed the consequences if it did.[9]

Table 2. Safety Measures and Unresolved Defects

Safety measure	Problem found after Flight 981
Reliable locking	The door could appear closed without being properly latched and locked.
Support plate	The support plate specified by Service Bulletin 52-37 was missing, although it should have been installed before delivery.
Correct adjustment	Lock-pin travel and the warning-switch adjustment were incorrect.
Visual inspection	The viewing-port check was not performed at Orly, and the port was too small for easy inspection.
Protection of the floor	Pressure relief was insufficient to prevent the floor damage that affected the controls.

Source:[10].

The FAA's account says it accepted a manufacturer-led modification program instead of issuing the proposed mandatory directive. However, inaccurate records of completed work also mattered: ordering a change would not be enough if nobody verified that it was installed.[11]

There were **629 days** between the two accidents. By then, the manufacturers and the FAA knew the door could fail, yet the aircraft remained in service without adequate repairs.

[8] NAT'L TRANSP. SAFETY BD., *supra* note 2, at Synopsis.

[9] NAT'L TRANSP. SAFETY BD., *supra* note 2, at app. G, 37–38.

[10] COMMISSION OF INQUIRY, *supra* note 1, at 29, 46–50.

[11] FED. AVIATION ADMIN., *McDonnell Douglas DC-10 TC-JAV*, at Common Themes, https://www.faa.gov/lessons_learned/transport_airplane/accidents/TC-JAV (last visited Sept. 18, 2026).

II.C. What happened on March 3, 1974

Flight 981 stopped at Paris-Orly before continuing to London. The aircraft, registration **TC-JAV**, was within its approved weight and balance limits. The investigation did not identify bad weather or overloading as the explanation for the crash.[12]

The following times are local to France. They follow the detailed flight history; some recordings and summaries differ slightly in timing.[13]

Table 3. Flight 981 Timeline

Local time	Event
11:02 a.m.	The aircraft arrived at Orly. Some passengers left and others boarded for London.
About 11:35 a.m.	The rear left cargo door was closed. Its outside appearance did not show that it was insecure.
12:24 p.m.	The crew received taxi clearance.
About 12:30 p.m.	The aircraft took off.
12:36 p.m.	The crew received clearance to continue climbing.
About 12:40 p.m.	The rear cargo door separated. Rapid decompression damaged the floor and controls.
About 77 seconds later	The aircraft struck the Ermenonville Forest near Fontaine-Chaalis.

Source:[14]. Times are local to France and approximate where indicated.

The door failed while the aircraft was climbing, at approximately 11,000–12,000 feet according to the investigation and official summaries. The crew then faced an aircraft whose flight controls had been severely damaged. This was not simply a warning light that could be addressed by following a normal checklist.[15,16]

All sources used here agree on **346 deaths**. They differ slightly in how they divide that total between passengers and crew, so the calculations use the total.[17,18]

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[12] COMMISSION OF INQUIRY, *supra* note 1, at 4–6, 46–48.

[13] COMMISSION OF INQUIRY, *supra* note 1, at 1 n.1; 4–6.

[14] COMMISSION OF INQUIRY, *supra* note 1, at 4–6.

[15] COMMISSION OF INQUIRY, *supra* note 1, at 43–48.

[16] *McDonnell Douglas DC-10 TC-JAV*, *supra* note 11, at History of Flight.

[17] COMMISSION OF INQUIRY, *supra* note 1, at 6.

II.D. How the door failure brought down the aircraft

The DC-10's cargo door opened outward. During flight, higher air pressure inside the aircraft pushed against it, so the latches and locks had to hold it securely. On Flight 981, the door could look closed even though the mechanism was not in its safe position. The warning system could also give a misleading indication.[19]

When the door separated, pressure in the cargo compartment dropped rapidly. Air in the passenger cabin could not escape downward quickly enough. The resulting pressure difference pushed down on the cabin floor and damaged it. Flight-control cables beneath the floor were affected, leaving the pilots unable to recover.[20]

Two protections were needed: a door that stayed secured and an aircraft that could survive a pressure loss. The accident exposed weaknesses in both.

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[18] *In re Paris Air Crash of March 3, 1974*, 399 F. Supp. 732, 735 (C.D. Cal. 1975).

[19] COMMISSION OF INQUIRY, *supra* note 1, at 34–36, 43.

[20] COMMISSION OF INQUIRY, *supra* note 1, at 44–48.

III. Parties and Causation

III.A. Plaintiffs: who could bring a claim?

The main plaintiffs would be the victims' eligible surviving relatives, seeking compensation for the losses caused by the deaths.

Table 4. Potential Plaintiffs and Claims

Possible plaintiff	What the claim could cover
Eligible surviving family members	Wrongful death: lost financial support, household services, companionship, and other legally recognized family losses.
An estate representative	A separate survival claim, which continues certain claims belonging to the person who died.
Crew members' families	Claims against responsible manufacturers; claims involving the airline may also involve employment rules.
Turkish Airlines	Damage to the aircraft and related losses claimed against the manufacturers. The airline could be both a plaintiff and a defendant in different claims.
Affected property owners	Property damage, if ownership and losses could be proved.

Sources:[21,22]. Claim eligibility depends on the applicable law and proof.

The 1975 court opinion described **203 suits involving 337 of the people who died**. It also recorded Turkish Airlines' claim for more than $35 million in aircraft-related losses and reimbursement from other defendants.[23]

Not every person close to a victim automatically had the right to sue. In the *Rankin* decision, the court rejected claims by plaintiffs described as "common-law wives" who had not alleged the qualifications required under the relevant California law.[24]

For the damages model, one claim means the combined eligible family losses associated with one death. The full amount should not be paid separately to each relative, because that would count the same lost income and services more than once.

[21] *In re Paris Air Crash*, 399 F. Supp. at 735–36.

[22] *In re Paris Air Crash of March 3, 1974*, 420 F. Supp. 880 (C.D. Cal. 1976).

[23] *In re Paris Air Crash*, 399 F. Supp. at 735–36.

[24] *In re Paris Air Crash*, 420 F. Supp. at 880–82.

III.B. Defendants: who contributed to the danger?

A **defendant** is a person or organization being sued. The main organizations in the actual litigation were McDonnell Douglas, General Dynamics, Turkish Airlines, and the United States. Their possible responsibility can be arranged from design decisions to the final checks before departure.[25]

Table 5. Potential Defendants and Their Roles

Actor	Possible responsibility
McDonnell Douglas	Responsible for the complete aircraft. The case would focus on unsafe design, incomplete corrections, and the aircraft's condition when delivered.
General Dynamics / Convair	Worked on the fuselage and cargo doors. Its design and manufacturing work, along with its knowledge of the danger, would need examination.
United States / FAA	Responsible for aviation oversight. The question is whether its response to known problems was adequate; government immunity presents a separate legal hell.
Turkish Airlines	Responsible for maintaining and operating the aircraft, including adjustments, procedures, and staff training.
Ground-service personnel and employer	Their role in closing and checking the door would need examination. A separate claim would require proof of their duties and employer.

Sources:[26,27,28].

The strongest argument against McDonnell Douglas is that it was responsible for how the door, floor, warning system, and controls worked together. Convair's early warnings establish its knowledge of the danger.[29]

The airline's maintenance and checking duties also mattered. However, blaming the last worker to touch the door would ignore the design that allowed it to appear safe when it was not. The investigation found both technical defects and failures in checking and correction.[30]

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[25] *In re Paris Air Crash*, 399 F. Supp. at 735–38.

[26] *In re Paris Air Crash*, 399 F. Supp. at 735–38.

[27] COMMISSION OF INQUIRY, *supra* note 1, at 46–50.

[28] SPECIAL SUBCOMM. ON INVESTIGATIONS, *supra* note 4, at 11–14.

[29] SPECIAL SUBCOMM. ON INVESTIGATIONS, *supra* note 4, at 11–14.

[30] COMMISSION OF INQUIRY, *supra* note 1, at 36, 46–50.

There is no established basis in this record for assigning causal fault to the passengers or blaming the pilots for failing to recover after the controls were damaged.

III.C. Actual and proximate causation

Actual causation asks whether an action or failure helped cause the injury. Here, a secure cargo door would have prevented the decompression sequence from starting. Better pressure relief and protection of the controls could also have reduced the consequences of losing the door.[31]

Proximate causation asks whether the harm was closely enough connected to the danger created to justify legal responsibility. A crash following door failure was a foreseeable kind of harm, especially after Flight 96 had already demonstrated damage to the floor and controls.[32]

The manufacturers could argue that maintenance mistakes or an omitted inspection caused the accident after delivery. The families could answer that the locking and warning systems were supposed to protect against foreseeable errors. California's *Stewart v. Cox* decision recognized that a later person's foreseeable negligence does not necessarily excuse an earlier wrong.[33]

My conclusion is that several failures contributed to the same crash. Their importance depends on what each organization knew and could have prevented, not just which person acted last like a game of hot potato.

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[31] COMMISSION OF INQUIRY, *supra* note 1, at 43–50.

[32] NAT'L TRANSP. SAFETY BD., *supra* note 2, at Synopsis, 1; app. G, 37–38.

[33] *Stewart v. Cox*, 55 Cal. 2d 857, 863–64 (1961).

IV. Jurisdiction

IV.A. Why California is the strongest choice

For the families, I would choose **California**, especially the **United States District Court for the Central District of California** if the requirements for federal jurisdiction were met. The strongest reason is practical: the actual crash litigation took place there, and California had an important connection to the aircraft's design and manufacture.[34]

Table 6. Reasons for Choosing California

Reason	Why it helps the plaintiffs
Manufacturing connection	The alleged design and manufacturing failures had a substantial connection to California.
Shared evidence	Many families needed the same company records, engineering evidence, and accident findings. Coordinating the cases reduced repeated work.
Actual litigation	The court handled a large group of Flight 981 cases, making this a historically supported choice.
California damages law	The court applied California law to the damages issues it considered.

Source:[35].

The 1975 opinion described 203 suits before the court. Coordinating cases helped resolve common issues while leaving each family to prove its losses.[36]

A California state court would be another possibility for suitable claims if a federal court could not hear them. Other states would need a specific connection to the parties or events; the existing record does not give a stronger reason to choose them.

California also had limits. Some relatives could not qualify to sue, and the later appellate ruling rejected punitive damages on the wrongful-death claims. A good choice of court must account for those disadvantages too.[37,38]

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[34] *In re Paris Air Crash of March 3, 1974*, 399 F. Supp. 732, 735–49 (C.D. Cal. 1975).

[35] *In re Paris Air Crash*, 399 F. Supp. at 735–49.

[36] *In re Paris Air Crash*, 399 F. Supp. at 735–36.

[37] *In re Paris Air Crash of March 3, 1974*, 420 F. Supp. 880 (C.D. Cal. 1976).

[38] *In re Paris Air Crash of March 3, 1974*, 622 F.2d 1315 (9th Cir. 1980).

IV.B. Limits on claims against the airline

Claims against Turkish Airlines require a different analysis from claims against the manufacturers. International airline travel was covered by the **Warsaw Convention** when its requirements were met. The passenger's full ticket and itinerary mattered, not just the Paris–London segment.[39]

Article 28(1) required suit in the territory of a High Contracting Party, before a court at the carrier's domicile, principal place of business, establishment through which the contract was made, or place of destination. A passenger's nationality alone did not create a United States forum.[40]

The treaty also included liability limits and exceptions. Their application depended on the governing version and travel arrangements. The 1975 court identified treaty questions as disputed and left some unresolved at that stage. Therefore, I have not applied one airline liability limit to all 346 deaths. A limit on the airline would not automatically limit a separate defective-product claim against a manufacturer.[41,42]

IV.C. The FAA: responsibility and federal immunity

The FAA warrants a separate liability analysis because it had an opportunity to interrupt a known sequence of catastrophic failure. The strongest historical fact is that **the United States was actually a defendant in the Flight 981 litigation**. In 1975, the district court rejected its argument that the crash's location in France barred the claims alleging negligent acts in California.[43]

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[39] *Convention for the Unification of Certain Rules Relating to International Carriage by Air* arts. 1, 17, Oct. 12, 1929, 49 Stat. 3000, https://www.mcgill.ca/iasl/files/iasl/warsaw1929.pdf.

[40] *Convention for the Unification of Certain Rules Relating to International Carriage by Air* art. 28(1), Oct. 12, 1929, 49 Stat. 3000, https://www.mcgill.ca/iasl/files/iasl/warsaw1929.pdf.

[41] *In re Paris Air Crash*, 399 F. Supp. at 737.

[42] *Convention for the Unification of Certain Rules Relating to International Carriage by Air* arts. 1, 17, 20–22, 25, Oct. 12, 1929, 49 Stat. 3000, https://www.mcgill.ca/iasl/files/iasl/warsaw1929.pdf.

[43] *In re Paris Air Crash*, 399 F. Supp. at 737–38.

This section distinguishes an argument available during the original litigation from a hypothetical claim assessed under later Supreme Court decisions. The distinction matters: subsequent cases strengthened defenses that a plaintiff in 1975 could still contest.

IV.C.1. What the FAA knew and what it did

The FAA's responsibilities included certification and oversight of compliance with aircraft safety standards. The NTSB investigated the 1972 American Airlines accident and recommended corrective measures; the FAA was the agency positioned to require regulatory action. The NTSB's July 6, 1972 recommendations, A-72-97 and A-72-98, addressed both the unreliable door-locking arrangement and inadequate pressure relief through cabin floors. These were separate protections against an accident sequence already demonstrated in service.[44]

Congress's December 1974 investigation supplies a particularly important account of the FAA response. Its Western Region staff prepared a mandatory airworthiness directive, or AD. However, after a June 15, 1972 telephone discussion between FAA Administrator John Shaffer and Douglas President Jackson McGowen, the agency pursued an informal arrangement with the manufacturer. The congressional report describes how regional director Arvin Basnight learned of that arrangement on June 16. This was a decision within the FAA hierarchy, not simply an inspector overlooking a part.[45]

The distinction between an AD and a service bulletin is substantive. An AD is a regulatory command within its applicable scope. A manufacturer's bulletin communicates a modification; calling it mandatory does not itself give it the legal force of an FAA directive. The FAA's retrospective account acknowledges that it accepted the manufacturer's approach and did not mandate incorporation of the three bulletins. TC-JAV's records indicated all three modifications were complete, although only two had been incorporated.[46]

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[44] NAT'L TRANSP. SAFETY BD., *supra* note 2, at app. G, 37–38.

[45] SPECIAL SUBCOMM. ON INVESTIGATIONS, *supra* note 4, at 7–9, 32–34.

[46] *McDonnell Douglas DC-10 TC-JAV*, *supra* note 11, at Prevailing Cultural / Organizational Factors.

The timing strengthens the criticism. The FAA issued a telegraphic AD on March 7, 1974, four days after the crash, and expanded the requirements on March 22. The directive's later consolidated text records both telegram dates.[47] Congress criticized the delay because important corrective measures had been available before the crash. The ethical question is why evidence sufficient to demand action after 346 deaths had not produced adequate intervention following the earlier near-disaster.[48]

IV.C.2. Causation: the strongest allegation and its weakness

The plaintiffs' theory would be that prompt mandatory corrections, effective communication, and verification before delivery or operation would have exposed the incomplete work and interrupted the failure sequence. Flight 96 makes a later crash from door separation and damage to controls foreseeably connected to the known defect. The French commission specifically criticized the absence of mandatory directives after that accident.[49]

However, **notice of a danger is not proof that the omitted regulatory action would have prevented this particular death**. Congress expressly declined to decide whether an AD would have saved TC-JAV. The missing modification was already recorded as installed. The government could therefore argue that a directive would have encountered the same false record. Congress also explained the plaintiffs' response: legal compulsion might have prompted more careful manufacturer verification, FAA inspection, or scrutiny by the Turkish authorities accepting delivery. Both possibilities require evidence.[50]

An FAA order also cannot simply be assumed to operate as a worldwide grounding command over a Turkish-registered aircraft in France. The proposed causal chain must identify its applicable reach and how corrective information or certification requirements would reach the manufacturer, airline, and foreign authorities. Useful evidence would include the proposed

[47] FED. AVIATION ADMIN., *Airworthiness Directive 74-08-04*, amend. 39-2443, at 4 (effective Dec. 1, 1975), https://www.faa.gov/sites/faa.gov/files/2022-10/AD74-08-04.pdf.

[48] SPECIAL SUBCOMM. ON INVESTIGATIONS, *supra* note 4, at 7–9.

[49] COMMISSION OF INQUIRY, *supra* note 1, at 49–50.

[50] SPECIAL SUBCOMM. ON INVESTIGATIONS, *supra* note 4, at 8–9.

directive, certification files, communications with Turkish officials, modification records, and the inspection requirements actually in force.

IV.C.3. How a damages suit against the government works

Sovereign immunity means that the United States can be sued only within a congressional waiver. The relevant route is the **Federal Tort Claims Act (FTCA)**. Under 28 U.S.C. § 1346(b)(1), federal district courts hear qualifying claims for harm caused by federal employees acting within their employment, where a private person would face liability under the applicable state law. The tort defendant is the **United States**, rather than the FAA in its own name. Section 2679(a) preserves that distinction for claims covered by the FTCA.[51,52]

Proving that the FAA violated federal aviation rules would not, by itself, establish liability under the FTCA. The plaintiffs must also show that state law would hold a private person or company liable for similar conduct. In *United States v. Olson*, 546 U.S. 43, 44–47 (2005), which involved federal mine inspectors, the Supreme Court required this comparison with a private party rather than a state or city regulator. For Flight 981, the relevant comparison could be a private company that agrees to inspect aircraft repairs but performs the inspection carelessly. Depending on the applicable state law, the plaintiffs would need to show how the inspector took responsibility for safety, caused others to rely on its work, or increased the danger through its negligence. The argument must connect the FAA's specific conduct to a duty recognized by state law; its general mission to promote aviation safety is not enough.[53]

[51] 28 U.S.C. § 1346(b)(1).

[52] 28 U.S.C. § 2679(a).

[53] *United States v. Olson*, 546 U.S. 43 (2005).

Table 7. Requirements for a Federal Tort Claim

Procedural issue	Effect on the proposed claim
Administrative claim	Present the claim to the appropriate federal agency first. It ordinarily must state a specific dollar amount. After a final written denial, or six months without final disposition, the claimant may proceed under § 2675(a).
Time limits	Section 2401(b) generally requires presentation within two years after accrual and suit within six months after mailing of a final denial. Accrual and any tolling require separate analysis.
Court and venue	The FTCA gives federal district courts jurisdiction. Section 1402(b) permits venue where the plaintiff resides or the complained-of act or omission occurred.
Trial and damages	The FTCA claim is tried to a judge. Section 2674 prohibits punitive damages and prejudgment interest against the United States.

Sources:[54,55,56,57,58,59,60].

For the historical case, California was supported by allegations of FAA acts there. Washington, D.C., would require its own venue basis, such as relevant acts there; the location of an agency's headquarters alone is insufficient. Venue does not remove immunity or automatically select the substantive damages law, nor would a federal claim automatically supply jurisdiction over every related private defendant. This analysis assumes a timely original lawsuit, not a newly filed claim decades after the accident.

Challenging agency action is also different from collecting wrongful-death damages. The current Administrative Procedure Act waiver in 5 U.S.C. § 702 concerns relief other than money damages; it is not a substitute compensation route around the FTCA's exceptions.[61]

[54] 28 U.S.C. § 1346(b)(1).

[55] 28 U.S.C. § 2675(a)–(b).

[56] 28 U.S.C. § 2401(b).

[57] 28 U.S.C. § 1402(b).

[58] 28 U.S.C. § 2402.

[59] 28 U.S.C. § 2674.

[60] 28 C.F.R. § 14.2(a) (2026).

[61] 5 U.S.C. § 702.

IV.C.4. The discretionary-function defense

Section 2680(a) preserves immunity for protected discretionary functions even when the discretion was abused. Thus, showing that a regulatory decision was unreasonable, dangerous, or influenced by industry does not by itself defeat the defense.[62]

United States v. Varig Airlines, 467 U.S. 797, 814–820 (1984), is especially damaging to the plaintiffs' certification theory. It concerned claims that federal aviation officials negligently certified aircraft with unsafe features. The Court protected the FAA's system of selective compliance review and employees' execution of that system. Its reasoning covered choices about inspection priorities and the use of limited resources. Applied here, the government would characterize certification oversight and the choice of corrective regulatory method as protected judgments. *Varig* did not decide Flight 981, but its reasoning would be powerful authority against a similar later claim.[63]

The narrower opening comes from *Berkovitz v. United States*, 486 U.S. 531, 536–548 (1988), concerning federal approval and release of a polio vaccine. Its analysis asks whether the employee had a choice and, if so, whether that choice involved the kind of policy judgment the exception protects. A specific, binding instruction that prescribes the employee's conduct can remove discretion. *Berkovitz* permitted claims to proceed on relevant allegations; it did not award damages or make every safety violation actionable.[64]

The strongest pleading would therefore identify a particular mandatory FAA step, the official required to perform it, and the evidence of nonperformance. A general instruction to promote safety is insufficiently specific. A requirement imposed on the manufacturer does not necessarily require an FAA inspector to personally verify every component. Most importantly, **a proposed AD that was never issued is evidence of an available response, not itself a binding command that an official violated**.

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[62] 28 U.S.C. § 2680(a).

[63] *United States v. S.A. Empresa de Viacao Aerea Rio Grandense (Varig Airlines)*, 467 U.S. 797 (1984).

[64] *Berkovitz v. United States*, 486 U.S. 531 (1988).

United States v. Gaubert, 499 U.S. 315, 322–325 (1991), prevents an easy workaround: calling conduct operational does not automatically remove immunity. Routine regulatory decisions can still involve protected policy judgment, and the inquiry concerns the nature of the act rather than an official's subjective thinking. That makes a broad allegation of careless follow-through weaker than proof of a specific nondiscretionary omission.[65]

The two steps both matter. In *Martin v. United States*, 605 U.S. 395 (2025), involving an FBI raid at the wrong home, the Court returned the discretionary-function question for further consideration. Justice Sotomayor's separate concurrence emphasized that absence of a mandatory instruction does not end the inquiry: the policy component still matters. That discussion is a concurrence, not an additional holding of the Court. *Martin* did not eliminate the defense or decide aircraft-certification liability.[66]

IV.C.5. The foreign-country defense: 1975 versus later law

The original *In re Paris Air Crash*, 399 F. Supp. 732, 737–738 (C.D. Cal. 1975), rejected the United States' § 2680(k) defense because the complaints alleged negligent acts in California. The court distinguished the place of the alleged wrong from the place of the accident. It also reported that the Ninth Circuit denied the government's mandamus request without an opinion.[67]

Later, *Sosa v. Alvarez-Machain*, 542 U.S. 692, 699–712 (2004), rejected a domestic-headquarters theory for injury suffered abroad. That case involved an abduction in Mexico planned by federal agents in the United States. The Court held that foreign-country injury remains within § 2680(k) despite domestic wrongful acts.[68]

Applied to an equivalent claim under that later rule, the deaths in France create a separate, likely decisive bar. Proving a nondiscretionary FAA violation would address § 2680(a), but would not remove § 2680(k).

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[65] *United States v. Gaubert*, 499 U.S. 315 (1991).

[66] *Martin v. United States*, 605 U.S. 395, 414–17 (2025).

[67] *In re Paris Air Crash of March 3, 1974*, 399 F. Supp. 732 (C.D. Cal. 1975).

[68] *Sosa v. Alvarez-Machain*, 542 U.S. 692 (2004).

IV.C.6. Precedent comparison

Table 8. Federal Immunity Precedents

Case	What it contributes, and its limit
Dalehite (1953)	After the Texas City fertilizer explosions, the Court protected challenged federal planning decisions. Government immunity was already a serious issue before Flight 981; it was not invented in 1984.
Indian Towing (1955)	A negligent lighthouse-maintenance claim could proceed against the United States. An undertaken safety service can create a duty of care. The government had conceded that the discretionary-function exception was not involved, limiting the analogy to regulatory choices.
Paris Air Crash (1975)	Direct historical support: the court rejected the foreign-country defense to the alleged California negligence. This was not a judgment establishing FAA fault.
Varig Airlines (1984)	Direct aviation authority protecting the challenged certification and selective-inspection process. Strong defense to broad FAA oversight allegations.
Berkovitz (1988) and *Gaubert* (1991)	Identify the boundary between mandatory conduct and policy discretion. A specific required act helps plaintiffs; merely labeling an error operational does not.
Sosa (2004)	Domestic planning does not avoid the foreign-country exception for injury abroad. This independently threatens an equivalent Flight 981 claim under later law.

Sources:[69,70,71,72,73,74,75].

Additional case citations: *Dalehite v. United States*, 346 U.S. 15, 35–42 (1953); *Indian Towing Co. v. United States*, 350 U.S. 61, 64–69 (1955).

Indian Towing provides a historically available answer to the claim that federal safety services are categorically immune. It supports a duty to exercise care after undertaking a service on which others rely. However, keeping an existing lighthouse working differs from deciding how to regulate a manufacturer.

[69] *Dalehite v. United States*, 346 U.S. 15 (1953).

[70] *Indian Towing Co. v. United States*, 350 U.S. 61 (1955).

[71] *In re Paris Air Crash of March 3, 1974*, 399 F. Supp. 732 (C.D. Cal. 1975).

[72] *United States v. S.A. Empresa de Viacao Aerea Rio Grandense (Varig Airlines)*, 467 U.S. 797 (1984).

[73] *Berkovitz v. United States*, 486 U.S. 531 (1988).

[74] *United States v. Gaubert*, 499 U.S. 315 (1991).

[75] *Sosa v. Alvarez-Machain*, 542 U.S. 692 (2004).

IV.C.7. Compensation and the FAA's responsibility

A successful FTCA claim would require the United States to compensate families for losses covered by the applicable state law. These can include lost financial support and companionship under California law.[76] A later explanation of the established California rule also confirms recovery of funeral expenses.[77] The damages section below explains the model. Federal law prohibits punitive damages against the United States, so the amount available to punish the government would be **$0**.[78]

The damages model includes the United States for the FAA's conduct. Its equal-barrier baseline assigns 37.5% to Douglas, 25% to General Dynamics, 25% to Turkish Airlines, and 12.5% to the United States. These are explicit classroom judgments about responsibility, not findings from the litigation. The model separately tests a successful FTCA claim, government immunity with full private collection, and a limited-collection stress test. The parties share responsibility for the same compensable losses; those losses are counted only once.

The FAA's ethical duty was to protect passengers through independent oversight. After Flight 96, it knew that a cargo-door failure could damage the cabin floor and flight controls. It should have required an effective correction, checked that the work was completed, and ensured that operators and foreign aviation authorities received clear safety instructions. Instead, the agency accepted an informal arrangement with the manufacturer. Congress sharply criticized that response and identified stronger verification as a possible benefit of a mandatory directive.[79,80]

The FAA failed to secure an adequate response to a known catastrophic danger. That is the central criticism supported by the investigation. The families' legal task was to turn that failure into a claim the FTCA allowed: identify the government's specific negligence, connect it to the crash, and overcome the immunity defenses discussed above. They made meaningful

[76] *Krouse v. Graham*, 19 Cal. 3d 59, 67–70 (1977).

[77] *Vander Lind v. Superior Ct.*, 146 Cal. App. 3d 358, 364 (1983).

[78] 28 U.S.C. § 2674.

[79] NAT'L TRANSP. SAFETY BD., *supra* note 2, at app. G.

[80] SPECIAL SUBCOMM. ON INVESTIGATIONS, *supra* note 4, at 7–9, 32–34.

progress in 1975 when the court rejected the government's argument that the crash's location in France barred their claims.[81,82]

[81] SPECIAL SUBCOMM. ON INVESTIGATIONS, *supra* note 4, at 7–9.

[82] *In re Paris Air Crash of March 3, 1974*, 399 F. Supp. 732, 737–38 (C.D. Cal. 1975).

V. Comparative Fault and Damages

V.A. Comparative negligence

Comparative negligence reduces compensation when an injured person's own negligence helped cause the injury. If a person had $100,000 in damages and was 20% responsible, a simple comparative-negligence calculation would leave $80,000.

$$\text{Recovery} = \text{Damages} \times (1 - \text{Plaintiff's fault share})$$

For Flight 981, there is no established causal fault by the passengers. The model therefore uses **0% passenger fault**, meaning their compensation is not reduced.

California's rules changed while the litigation was underway:

Table 9. Changes in California Comparative-Fault Law

Year	Decision	Main point
1975	*Li v. Yellow Cab*	California adopted pure comparative negligence, with rules limiting how it applied to pending cases.
1978	*Daly v. General Motors*	Comparative-fault principles were applied to strict product-liability cases.
1978	*American Motorcycle*	The court allowed responsibility to be shared among defendants while preserving joint and several liability.

Sources:[83,84,85].

Sharing responsibility among defendants is different from reducing a passenger's compensation. If the airline is assigned 25% of the responsibility, that does not mean a passenger loses 25% of the award. The airline's negligence is not the passenger's negligence.

Joint and several liability can allow an injured person to collect the full covered loss from a responsible defendant, with the defendants resolving reimbursement among themselves. The family cannot collect the same loss twice.[86]

[83] *Li v. Yellow Cab Co.*, 13 Cal. 3d 804 (1975).

[84] *Daly v. Gen. Motors Corp.*, 20 Cal. 3d 725 (1978).

[85] *Am. Motorcycle Ass'n v. Superior Ct.*, 20 Cal. 3d 578 (1978).

[86] *Am. Motorcycle Ass'n v. Superior Ct.*, 20 Cal. 3d 578 (1978).

Later in this report, a four-party safety-barrier model derives a 37.5% / 25% / 25% / 12.5% reference allocation and tests both factual attribution and legal collectibility. McDonnell Douglas, General Dynamics, and Turkish Airlines formalized a confidential agreement to share the plaintiffs' damages.[87]

V.B. Economic damages: assumptions

Economic damages include lost family support and household services.[88] California's former wrongful-death statute also allowed funeral expenses, as a later opinion explains.[89] To estimate lost support, this model deducts the assumed portion of income and benefits consumed personally by the decedent.

The example below assumes an adult who would have contributed to the household for another 30 years. Annual income is set at **$5,335**, the U.S. median personal income in 1974 reported by the Census Bureau. This is an individual income measure, not household income or the median salary of full-time workers. All amounts remain in 1974 dollars.[90]

[87] *In re Paris Air Crash*, 399 F. Supp. at 738.

[88] *Krouse v. Graham*, 19 Cal. 3d 59, 67–70 (1977).

[89] *Vander Lind v. Superior Ct.*, 146 Cal. App. 3d 358, 364 (1983).

[90] U.S. CENSUS BUREAU, *Median Personal Income in the United States (MEPAINUSA646N)*, FED. RSRV. BANK OF ST. LOUIS, https://fred.stlouisfed.org/data/MEPAINUSA646N (last visited Sept. 18, 2026).

Table 10. Baseline Economic-Loss Assumptions

Assumption	Amount
Annual personal income	$5,335
Annual benefits contributing to family support	$1,500
Share spent on personal needs	25%
Annual value of household services	$2,000
Remaining years of contributions	30 years
Real annual discount rate	3%
Real annual growth in contributions	0%
Funeral and burial expense	$2,000

Source: income from the U.S. Census Bureau;[91] remaining inputs are author-selected assumptions.

Each entry represents part of the support the family would lose or an assumption about how to value it. The income figure comes from Census data; the other values are estimates chosen for this example.

Annual personal income — $5,335. This is the person's yearly income before taxes. Personal income can include wages and other money income. The model uses the national median as a starting point; a calculation for a particular victim would use that person's income records.[92,93]

Annual benefits — $1,500. This represents additional benefits that helped support the family, such as employer-provided family health coverage. Census money income excludes noncash benefits.[94] Only additional family benefits should be added, to avoid counting the same support twice.

Personal needs — 25%. The person would have spent some of their income on their own food, clothing, and other needs. Because the calculation estimates the surviving family's lost

[91] U.S. Census Bureau, *Median Personal Income*, *supra* note 90.

[92] U.S. Census Bureau, *Median Personal Income*, *supra* note 90.

[93] U.S. CENSUS BUREAU, *About Income*, https://www.census.gov/topics/income-poverty/income/about.html (last visited Sept. 18, 2026).

[94] *About Income*, *supra* note 93.

support, it subtracts this personal spending. The model assumes that 25% of income and benefits went toward personal needs, leaving 75% for the family.

Household services — $2,000 per year. Cooking, cleaning, childcare, and home repairs have economic value even when nobody is paid to do them. The family might have to pay someone to replace that work. This entry estimates the annual value of the victim's unpaid work for other household members, separately from the money they provided.

Remaining contributions — 30 years. The example assumes the person would have continued providing financial support and household services for another 30 years. Age, health, retirement plans, and family circumstances would affect the period used for each person. Household work could also continue after retirement from paid employment.

Real annual discount rate — 3%. The family would have received support gradually, but compensation can be paid as one lump sum. Money received immediately can earn a return before it is needed, so the amount paid now can be smaller than the sum of all future payments. For example, at a 3% annual return, about $97.09 today would grow to $100 in one year. Discounting applies that reasoning to each year's loss. The assumed rate is a *real* rate, meaning it accounts for returns after inflation; it is consistent with expressing the losses in 1974 dollars.

Real annual growth — 0%. The model assumes the person's annual contributions would keep the same purchasing power over time. Income could rise with inflation, but the example includes no additional growth from promotions or raises above inflation. This is why the same annual contribution is used for all 30 years.

Funeral and burial expense — $2,000. This is a one-time cost rather than an annual loss. The model treats it as an immediate expense, so it is added once after future support has been discounted. An actual claim would use documented expenses.

First, subtract personal spending from income and benefits. Then add the value of household services:

$$A = (5,335 + 1,500) \times 0.75 + 2,000 = 7,126.25$$

The model therefore assumes an annual family loss of **$7,126.25**: $5,126.25 in financial support plus $2,000 in household services. The funeral expense is added separately in the next calculation.

Individual claims would require adjustments for taxes, employment history, and changing family needs.

The result also depends heavily on the assumptions. Keeping the other inputs fixed, using 20 years instead of 30 gives **$108,020.61** in economic damages; using 40 years gives **$166,721.64**.

V.C. Economic damages: calculation

The model values 30 years of annual contributions at the time of the accident, then adds the immediate funeral expense. The formula below assumes the contributions arrive at the end of each year:

$$E = 7,126.25 \times \frac{1 - 1.03^{-30}}{0.03} + 2,000$$

The present-value factor is **19.60044**. Components and totals below are independently rounded.

Table 11. Present Value of Economic Losses

Component	Present value
Financial support: $5,126.25 × 19.60044	$100,476.76
Household services: $2,000 × 19.60044	$39,200.88
Funeral expense	$2,000.00
Total economic damages	**$141,677.65**

Source: author calculations using the stated baseline assumptions. Components rounded independently.

Without discounting, the contributions and funeral expense total **$215,787.50**. Discounting gives **$141,677.65** because a lump sum can earn a return before the family needs it. A higher assumed rate therefore produces a lower present value, as the figure shows.

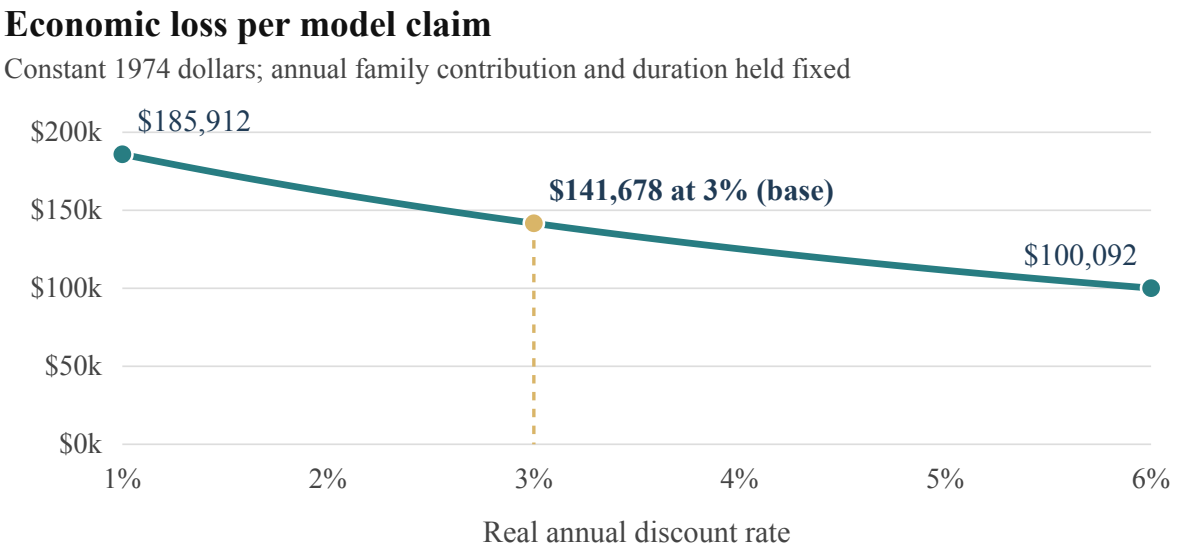


Figure 1: Only the real discount rate varies; annual loss remains $7,126.25 for 30 years, plus $2,000 funeral expense. The gold marker identifies the 3% baseline.

V.D. Noneconomic damages and total compensation

Noneconomic damages cover recognized losses that do not have an ordinary market price, such as companionship, care, and parental guidance. California's *Krouse v. Graham* decision distinguished those relationship losses from grief and sorrow alone, which were not separately recoverable in the wrongful-death action.[95]

For the example, I use **$100,000 in noneconomic damages** for the eligible family members' combined loss associated with one death. There is no formula that converts a relationship into a dollar amount, unless that amount happens to be zero.

Economic and noneconomic damages together are **compensatory damages**: both aim to compensate for loss.

$$C = 141,677.65 + 100,000 = 241,677.65$$

[95] *Krouse v. Graham*, 19 Cal. 3d 59, 67–70 (1977).

Table 12. Compensation per Assumed Family Claim

Example claim	Amount
Economic damages	$141,677.65
Noneconomic damages	$100,000.00
Total compensation	**$241,677.65**

Source: author calculations and assumed noneconomic loss.

To show the overall scale, the next table applies the same example to all 346 deaths. We round only after calculating the totals, so multiplying the rounded amounts shown above gives a slightly different result.

Table 13. Illustrative Compensation for 346 Deaths

Illustrative total for 346 deaths	Amount
Economic damages	$49,020,465.23
Noneconomic damages	$34,600,000.00
Total compensation	**$83,620,465.23**

Source: author calculations for 346 identical assumed claims. Totals calculated before rounding.

V.E. Allocating responsibility: four parties, four safeguards

The model allocates responsibility among **McDonnell Douglas, General Dynamics/ Convair, Turkish Airlines, and the United States for the FAA's conduct** using four safety barriers. The FAA's oversight and the manufacturer's verification duties share the corrective-action barrier because they address the same preventive opportunity: ensuring that known defects were effectively corrected.

Table 14. Four Safeguards and Preventive Opportunities

Barrier	Evidence and preventive action
A. Safe door design	A reliable interlock would prevent an unsafe closure from appearing flight-ready. Douglas and Convair controlled the design.
B. Decompression protection	Adequate pressure relief and protection of controls would interrupt the door-loss, floor-failure, and control-loss sequence.
C. Verified retrofit	The support plate was missing despite installation records. Manufacturer verification and an effective regulatory response to the known defect offered overlapping corrective opportunities.
D. Operating safeguards	Correct adjustment, compliant maintenance, training, and the viewing-port check could address the unsafe operating condition.

Sources:[96,97].

The accident report supports the physical barriers; Congress documents the manufacturers' roles and the FAA's decision to pursue an informal correction program.[98,99]

Evidence identifies the failures; judgment assigns the fractions. Neither the reports nor the 1975 ruling establish a numerical FAA fault share. In particular, Congress did not determine that an AD would have prevented this crash. This model assumes some effective FAA intervention would have improved verification, then tests how much responsibility that warrants.[100]

V.F. Who controlled each safeguard?

Let M mean Douglas, G mean General Dynamics, T mean Turkish Airlines, and U mean United States/FAA. Each row sums to one. The baseline assigns half of barrier C to Douglas and half to the FAA.

[96] COMMISSION OF INQUIRY, *supra* note 1, at 46–50.

[97] SPECIAL SUBCOMM. ON INVESTIGATIONS, *supra* note 4, at 7–14, 32–34.

[98] COMMISSION OF INQUIRY, *supra* note 1, at 46–50.

[99] SPECIAL SUBCOMM. ON INVESTIGATIONS, *supra* note 4, at 7–14, 32–34.

[100] SPECIAL SUBCOMM. ON INVESTIGATIONS, *supra* note 4, at 8–9.

Table 15. Baseline Responsibility by Safeguard

Barrier	M	G	T	U
A. Safe door design	1/2	1/2	0	0
B. Decompression protection	1/2	1/2	0	0
C. Verified retrofit	1/2	0	0	1/2
D. Operating safeguards	0	0	1	0
Responsibility units	**1.5**	**1**	**1**	**0.5**

Source: author-selected allocations based on the evidence discussed below. These are not adjudicated fault shares.

A and B: Douglas directed the overall design; Convair performed detailed fuselage and door work and identified decompression risk. Equal division is a modeling choice, not a finding of equal legal fault.[101]

C: Douglas's delivery records and the missing modification support manufacturer responsibility. The FAA knew the earlier failure sequence and accepted a manufacturer-led program without the proposed mandatory directive. The 50/50 division gives both corrective roles weight without assuming the FAA physically installed the part.[102,103]

D: Turkish Airlines bears the operator-level duty in this simplified model. Maintenance, instructions, and ground-service contracts could justify reallocating some responsibility to a manufacturer or contractor.[104]

The FAA receives no separate A, B, or D units. This deliberately concentrates its attribution on the strongest developed theory: an inadequate response after the earlier warning. It does not decide that certification oversight was faultless.

V.G. Calculating the four-party shares

For party i, let $a_{b,i}$ be its fraction of barrier b and w_b the barrier's importance weight:

[101] SPECIAL SUBCOMM. ON INVESTIGATIONS, *supra* note 4, at 11–14.

[102] *McDonnell Douglas DC-10 TC-JAV*, *supra* note 11, at History of Flight.

[103] SPECIAL SUBCOMM. ON INVESTIGATIONS, *supra* note 4, at 7–9.

[104] COMMISSION OF INQUIRY, *supra* note 1, at 36, 46–50.

$$s_i = \frac{\sum_{b=1}^{4} w_b a_{b,i}}{\sum_{b=1}^{4} w_b}$$

With all four weights equal to 1, Douglas receives $\frac{1.5}{4} = 37.5\%$; General Dynamics and Turkish Airlines each receive $\frac{1}{4} = 25\%$; the United States receives $\frac{0.5}{4} = 12.5\%$.

Allocation across 346 assumed claims:

Table 16. Compensation Allocated Across 346 Assumed Claims

Party	Share	Economic	Noneconomic	Total
McDonnell Douglas	37.5%	$18,382,674.46	$12,975,000.00	$31,357,674.46
General Dynamics	25.0%	$12,255,116.31	$8,650,000.00	$20,905,116.31
Turkish Airlines	25.0%	$12,255,116.31	$8,650,000.00	$20,905,116.31
United States / FAA	12.5%	$6,127,558.15	$4,325,000.00	$10,452,558.15
Total	**100%**	$49,020,465.23	$34,600,000.00	$83,620,465.23

Source: author calculations using baseline fault shares. Allocation does not establish collectibility.

Each economic allocation is $s_i \times 49,020,465.23$ and each noneconomic allocation is $s_i \times 34,600,000$, using the unrounded economic total internally. For the United States, the combined allocation is approximately 0.125×83.620465 million, or **$10.45 million**.

The families' combined compensatory loss is **$83,620,465.23**. The four-party allocation divides responsibility for that loss without duplicating compensation. The model assumes no decedent comparative fault, including for the crew. Independently rounded cells may differ from their displayed sum by one cent. All calculations retain full precision until presentation.

V.H. Individual claims and different loss assumptions

The same shares apply to the example family's combined claim. Real victims' families would need individualized earnings, support, services, and relationship-loss evidence.

Table 17. Compensation Allocated per Assumed Family Claim

Party	Economic	Noneconomic	Total
McDonnell Douglas	$53,129.12	$37,500.00	$90,629.12
General Dynamics	$35,419.41	$25,000.00	$60,419.41
Turkish Airlines	$35,419.41	$25,000.00	$60,419.41
United States / FAA	$17,709.71	$12,500.00	$30,209.71

Source: author calculations using baseline fault shares.

The United States' illustrative share is about **$30,209.71 per model claim**: $17,709.71 economic and $12,500 noneconomic. The latter is compensatory loss of relationship, not punishment merely because it lacks a market price. Recoverability still depends on applicable law and a valid FTCA claim.[105,106]

Loss assumptions and fault assumptions are separate variables. Applying the 12.5% federal share to lower, base, and upper loss scenarios gives:

Table 18. Loss Scenarios and Federal Allocation

Loss inputs	All claims	United States	Private parties
Lower	$57,488,707.96	$7,186,088.50	$50,302,619.47
Base	$83,620,465.23	$10,452,558.15	$73,167,907.07
Upper	$354,258,034.89	$44,282,254.36	$309,975,780.53

Source: author calculations using the lower, base, and upper assumptions below.

Every row uses 346 identical assumed claims. Differences in total compensation reflect the assumed losses; the number of defendants does not determine the value of a claim. Neither the FAA attribution nor the lower/upper loss cases represent a statistical confidence interval.

V.I. Loss scenario assumptions

The preceding lower, base, and upper totals use the inputs below for each of 346 identical assumed claims. All money amounts are in 1974 dollars. Except for the base income figure, these

[105] *Krouse v. Graham*, 19 Cal. 3d 59 (1977).

[106] 28 U.S.C. § 2674.

inputs are author-selected assumptions.[107]

Table 19. Inputs for the Three Loss Scenarios

Input per claim	Lower	Base	Upper
Annual income	$10,000	$5,335	$25,000
Annual benefits	$1,000	$1,500	$2,500
Personal consumption	30%	25%	20%
Annual household services	$1,500	$2,000	$3,000
Years of contributions	20	30	40
Real discount rate	5%	3%	1%
Real annual growth	0%	0%	0%
Funeral expense	$1,500	$2,000	$3,000
Noneconomic damages	$50,000	$100,000	$200,000

Source: base income from the U.S. Census Bureau;[108] other inputs are author-selected assumptions.

For each case, annual loss equals income plus benefits, less personal consumption, plus household services. Discount that annual loss over the stated period, then add funeral expense and noneconomic damages. Multiply the result by 346; the federal allocation is 12.5% of that total. "Lower" describes the total loss, not every input. Its income exceeds the base income, but its shorter period, higher discount rate, and smaller noneconomic allowance produce a lower total.

V.J. How sensitive is the responsibility allocation?

The baseline gives the FAA half of barrier C. Call that fraction q. At equal weights, the four shares are:

$$s_M = \frac{2 - q}{4}, \quad s_G = \frac{1}{4}, \quad s_T = \frac{1}{4}, \quad s_U = \frac{q}{4}$$

Lower FAA attribution uses $q = 0.25$; higher attribution uses $q = 0.75$. The zero-attribution case assigns responsibility solely to the three companies.

[107]U.S. Census Bureau, *Median Personal Income*, *supra* note 90.

[108]U.S. Census Bureau, *Median Personal Income*, *supra* note 90.

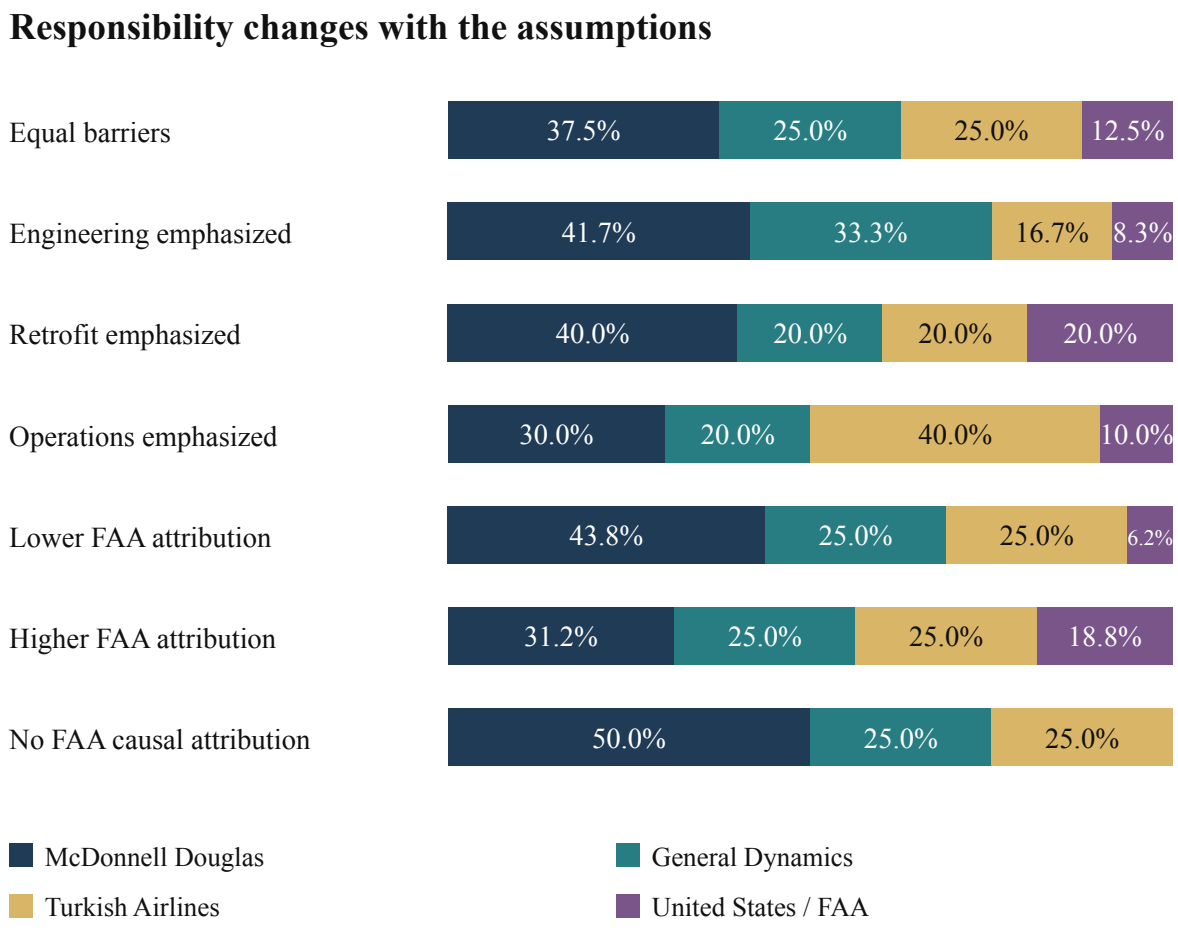


Figure 2: Engineering doubles A and B; retrofit doubles C; operations doubles D. Other rows vary only the FAA's fraction of C. All allocations sum to 100% before rounding.

V.K. What the sensitivity results establish

At equal weights, lower FAA attribution produces **43.75% / 25% / 25% / 6.25%**; higher attribution produces **31.25% / 25% / 25% / 18.75%**. No FAA causal attribution produces **50% / 25% / 25% / 0%**.

For the weights-only test, each barrier independently ranges from 1 to 2 while $q = 0.5$. The combined sensitivity test also varies q from 0.25 to 0.75. The analysis evaluates all 16 weight corners and all 32 combined corners.

Table 20. Sensitivity of Responsibility Shares

Party	Weights only	Weights + attribution
McDonnell Douglas	30.0%–42.9%	25.0%–50.0%
General Dynamics	16.7%–33.3%	16.7%–33.3%
Turkish Airlines	14.3%–40.0%	14.3%–40.0%
United States / FAA	7.1%–20.0%	3.6%–30.0%

Source: author calculations varying safeguard weights and FAA attribution. Ranges are conditional, not confidence intervals.

Each party's extremes may arise under different assumptions; **do not add the endpoints together**. These ranges remain conditional on the rest of the duty matrix. They omit scenarios such as a larger share for Douglas's design authority or an additional maintenance contractor.

The most useful new evidence would identify the proposed AD's reach, the mandatory inspection duties actually in force, who verified the missing plate, and how Turkish authorities would have responded. Those facts bear on whether FAA intervention would have prevented the crash. Legal immunity is a separate issue.[109]

V.L. Government immunity: who actually pays?

Responsibility allocation and collection are separate steps.

1. Federal claim succeeds. Payments follow the classroom shares, conditional on timely FTCA compliance, a state-law private-person analogue, proven causation, and no applicable immunity. This does not establish that plaintiffs overcame those defenses.[110,111]

2. Federal immunity; full private collection. Assume the private defendants remain liable for the indivisible loss, applicable joint and several liability permits full recovery, and they can pay. Their illustrative internal division is normalized to $\frac{37.5}{87.5}$, $\frac{25}{87.5}$, and $\frac{25}{87.5}$, or **42.857% / 28.571% / 28.571%**. Families recover the full total once.[112]

[109] SPECIAL SUBCOMM. ON INVESTIGATIONS, *supra* note 4, at 8–9.

[110] 28 U.S.C. § 2674.

[111] 28 U.S.C. § 1346(b)(1).

[112] *Am. Motorcycle Ass'n v. Superior Ct.*, 20 Cal. 3d 578 (1978).

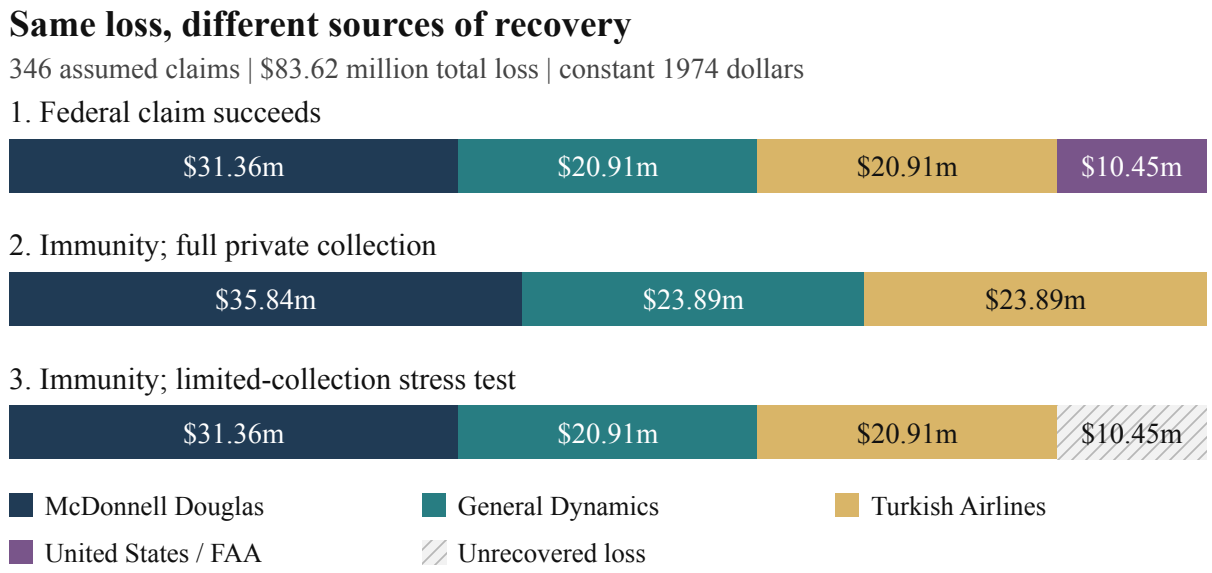


Figure 3: Author calculations: each bar represents $83.62 million in losses. Colors show payments; hatching shows unrecovered loss. Immunity alone does not establish the stress-test shortfall.

3. Limited-collection stress test. Assume, only to expose the financial risk, that private payments stop at their baseline responsibility shares and no one covers the federal allocation. Recovery is 87.5%, leaving a 12.5% gap.

V.M. Interpreting the federal recovery scenarios

Table 21. Payments Under Federal Recovery Scenarios

Party	Federal claim succeeds	Immunity; full collection	Immunity; stress test
McDonnell Douglas	$31,357,674.46	$35,837,342.24	$31,357,674.46
General Dynamics	$20,905,116.31	$23,891,561.49	$20,905,116.31
Turkish Airlines	$20,905,116.31	$23,891,561.49	$20,905,116.31
United States / FAA	$10,452,558.15	$0.00	$0.00
Total recovered	$83,620,465.23	$83,620,465.23	$73,167,907.07
Unrecovered loss	$0.00	$0.00	$10,452,558.15

Source: author calculations under the stated collection assumptions.

Government nonpayment can follow either unproven FAA causation or immunity despite an assumed causal contribution. The model distinguishes those explanations.

Under the later doctrine discussed in the FAA section, *Sosa* presents a likely decisive foreign-country bar, and *Varig* strongly protects the challenged certification and compliance-review choices. The successful federal-payment scenario is therefore not the default prediction for a comparable claim under that later law. The 1975 ruling must be assessed in its own historical setting.[113,114,115]

The full-collection scenario applies the historical joint-and-several framework discussed earlier. It does not import later changes in California law, resolve treaty limits affecting the airline, or establish a right of reimbursement against an immune government. Each private defendant must first be legally liable for the covered injury; collectibility, settlements, and any applicable limitations require separate analysis.

Numerical consequence: the $10,452,558.15 federal allocation is paid by private defendants under full collection or remains unrecovered under the stress test. Either way, the underlying compensatory loss remains $83,620,465.23.

McDonnell Douglas, General Dynamics, and Turkish Airlines formalized a confidential agreement to share damages.[116]

V.N. Punitive damages

Punitive damages punish particularly blameworthy conduct and discourage similar conduct. They are different from compensation. Evidence that a company knew of a serious danger would matter, but serious wrongdoing alone does not make punitive damages available under every type of claim.

For the historical California wrongful-death claims reviewed on appeal, the answer is **$0 in punitive damages**. In 1980, the Ninth Circuit reversed the earlier ruling that had allowed those punitive claims to proceed.[117]

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[113] *In re Paris Air Crash of March 3, 1974*, 399 F. Supp. 732 (C.D. Cal. 1975).

[114] *United States v. S.A. Empresa de Viacao Aerea Rio Grandense (Varig Airlines)*, 467 U.S. 797 (1984).

[115] *Sosa v. Alvarez-Machain*, 542 U.S. 692 (2004).

[116] *In re Paris Air Crash*, 399 F. Supp. at 738.

[117] *In re Paris Air Crash of March 3, 1974*, 622 F.2d 1315, 1316–23 (9th Cir. 1980).

The court distinguished wrongful-death claims from survival claims, which preserve certain claims of the person who died. It noted that the plaintiffs had agreed to give up their opportunity to pursue punitive damages through the survival route discussed in the opinion.[118]

Federal law also prohibits punitive damages against the United States under the Federal Tort Claims Act.[119]

Table 22. Compensation and Historical Punitive-Damages Result

Historical-law classroom model	Amount
Compensation across 346 assumed claims	$83,620,465.23
Wrongful-death punitive damages	$0.00
Combined total	**$83,620,465.23**

Sources: author compensation model.[120,121]

Historical-model result: punitive damages are zero for each private defendant under the wrongful-death ruling and zero for the United States under the FTCA. The baseline allocation changes neither rule. The next subsection separates hypothetical private-party punishment from these historical results.

V.O. Punitive sensitivity: private defendants only

For a different, legally valid claim allowing punitive damages, this hypothetical applies ratios of 0.5:1, 1:1, and 2:1 to each private defendant's allocated compensation. The United States remains at **$0 in every case**.[122]

$$P_i = k_i \times (s_i \times C), \quad P_U = 0$$

Here C is the full compensatory loss and k_i is a chosen illustrative ratio. The table gives all three private defendants the same ratio solely for comparison. Actual entitlement and amount would require separate proof about each defendant and compliance with applicable punitive-damages law; fault percentage alone cannot establish them.

[118] *In re Paris Air Crash*, 622 F.2d at 1316 n.1.

[119] 28 U.S.C. § 2674.

[120] *In re Paris Air Crash of March 3, 1974*, 622 F.2d 1315 (9th Cir. 1980).

[121] 28 U.S.C. § 2674.

[122] 28 U.S.C. § 2674.

Table 23. Hypothetical Private-Party Punitive Damages

Party / total	0.5:1	1:1	2:1
McDonnell Douglas	$15,678,837.23	$31,357,674.46	$62,715,348.92
General Dynamics	$10,452,558.15	$20,905,116.31	$41,810,232.61
Turkish Airlines	$10,452,558.15	$20,905,116.31	$41,810,232.61
United States / FAA	$0.00	$0.00	$0.00
Total punitive	$36,583,953.54	$73,167,907.07	$146,335,814.15
Compensation + punitive	$120,204,418.76	$156,788,372.30	$229,956,279.38

Source: author calculations at illustrative ratios.

At 1:1, hypothetical private punitive damages total **$73,167,907.07**, or 87.5% of compensation. Combined compensation and hypothetical punitive damages equal **$156,788,372.30**. The example family's equivalent 1:1 figures are **$211,467.94** hypothetical punitive damages plus **$241,677.65** compensation, totaling **$453,145.58**, rounded from unrounded amounts. These calculations use the baseline allocated private compensation even if collection later shifts under joint and several liability.

VI. Ethical Obligations

The defendants' ethical responsibilities went beyond avoiding a lawsuit. Passengers depended on organizations with information and technical expertise that they could not check for themselves. After the 1972 accident, those organizations had a strong reason to act before the same danger caused a worse outcome.

Table 24. Ethical Obligations by Organization

Organization	What it should have done
McDonnell Douglas	Correct the door problem, test the effects of decompression, and physically verify that required modifications were installed.
General Dynamics / Convair	Communicate design risks clearly, correct problems within its work, and keep pressing unresolved safety concerns.
Turkish Airlines	Maintain accurate records, check aircraft condition, and train staff to perform practical, understandable door inspections.
FAA	Respond firmly to a known catastrophic risk and verify that corrective action reached the affected aircraft.

Sources:[123,124].

These judgments follow the documented failures and recommendations in the accident report and congressional investigation.[125,126]

The missing support plate shows why paperwork alone was not enough. A signed record cannot protect passengers if the part is absent. Likewise, an inspection procedure is weak if the viewing port makes the mechanism difficult to see. A safety measure has to work on the actual aircraft, under the conditions in which people use it.

After the crash, the organizations also owed families honest information, preservation of evidence, and fair handling of claims. Legal limits on compensation do not reduce the human importance of those obligations.

Flight 981 was preventable. The four-party model distinguishes responsibility from collectibility. Federal immunity may change who pays; it does not erase the families' losses. Historical wrongful-death punitive damages remain zero.

[123] COMMISSION OF INQUIRY, *supra* note 1, at 46–50.

[124] SPECIAL SUBCOMM. ON INVESTIGATIONS, *supra* note 4, at 7–14, 32–34.

[125] COMMISSION OF INQUIRY, *supra* note 1, at 46–50.

[126] SPECIAL SUBCOMM. ON INVESTIGATIONS, *supra* note 4, at 7–14, 32–34.

VII. Conclusion

I would focus the families' case on McDonnell Douglas, General Dynamics/Convair, and Turkish Airlines, whose design, corrective-action, and operating responsibilities contributed to the failure sequence. California is the strongest supported U.S. forum, subject to the jurisdictional requirements for each claim. The earlier warnings strengthen the argument that this disaster was foreseeable. The FAA's response also deserves scrutiny, although overcoming federal immunity is a separate type of hell.[127,128]

The model illustrates how compensation and responsibility could be calculated. Its baseline gives $241,677.65 in compensation per assumed family claim and $83,620,465.23 across 346 identical assumed claims, with no reduction for passenger fault. Historical wrongful-death punitive damages remain zero.[129] The central ethical failure was allowing a known danger to remain inadequately corrected. Each organization had to make its safety measures work in practice, and the families' ability to collect damages is distinct from that obligation.

[127] COMMISSION OF INQUIRY, *supra* note 1, at 46–50.

[128] *In re Paris Air Crash of March 3, 1974*, 399 F. Supp. 732, 735–49 (C.D. Cal. 1975).

[129] *In re Paris Air Crash of March 3, 1974*, 622 F.2d 1315 (9th Cir. 1980).

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